

Project – 3

COVID-19 Data Analysis

Countries Analysed	India, USA, Brazil, UK, Germany
Date Range	Jan 2020 – Aug 2024
Data Source	Our World in Data (OWID)
Charts Generated	6 Publication-Quality Figures
Tools Used	Python · Pandas · NumPy · Matplotlib · SciPy · ReportLab

This report delivers an end-to-end epidemiological analysis of COVID-19 using real-world time-series data. It covers data ingestion, daily/weekly case computation, smoothing via rolling averages, peak wave detection with SciPy, reproduction-rate (R) trend analysis, and key cross-country comparative insights — all accompanied by production-quality charts and source code.

1. Data Loading & Feature Engineering

We sourced the OWID consolidated COVID-19 dataset — over 429,000 records covering 200+ countries — and filtered to 5 representative nations. Key engineered features include 7-day rolling average (to smooth reporting noise) and 7-day rolling sum (as a weekly case proxy).

Source Code – Data Ingestion & Feature Engineering

```
import pandas as pd, numpy as np # Load OWID time-series dataset url =
"https://raw.githubusercontent.com/owid/covid-19-data/master/public/data/owid-covid-data.csv"
df_raw = pd.read_csv(url, parse_dates=['date']) # Filter countries of interest COUNTRIES =
['India', 'United States', 'Brazil', 'United Kingdom', 'Germany'] cols =
['location', 'date', 'new_cases', 'total_cases', 'new_deaths',
'total_deaths', 'reproduction_rate', 'new_cases_smoothed'] df =
df_raw[df_raw['location'].isin(COUNTRIES)][cols].copy() df.sort_values(['location', 'date'],
inplace=True) # Feature Engineering # 7-day rolling average (smooths daily reporting noise)
df['rolling_7d'] = df.groupby('location')['new_cases'] \ .transform(lambda x: x.rolling(7,
min_periods=1).mean()) # 7-day rolling sum (proxy for weekly incidence) df['weekly_cases'] =
df.groupby('location')['new_cases'] \ .transform(lambda x: x.rolling(7, min_periods=1).sum())
print(f"Date range: {df['date'].min().date()} to {df['date'].max().date()}") print(f"Total rows:
{len(df):,}") # Output: 8,378
```

Interpretation

- **8,378 rows** across 5 countries spanning Jan 2020 – Aug 2024 (4.5+ years).
- Raw daily case counts exhibit strong weekly cycles (weekend under-reporting) — rolling average is essential.
- The **7-day rolling mean** (rolling_7d) removes noise without introducing too much lag.
- The **7-day rolling sum** (weekly_cases) approximates weekly incidence — a standard WHO metric.

2. Country Comparisons & Rolling Averages

Three visualisations are produced: per-country bar+line combo charts, an overlay of all rolling averages, and a weekly case trend. Together they reveal timing, magnitude, and relative severity of each nation's waves.

Source Code – Plotting

```
import matplotlib.pyplot as plt, matplotlib.dates as mdates colors_map = {'India':'#e63946','United States':'#457b9d','Brazil':'#2a9d8f', 'United Kingdom':'#e9c46a','Germany':'#f4a261'} #
Per-country: daily bars + 7-day rolling line fig, axes = plt.subplots(5, 1, figsize=(14, 18),
sharex=False) for ax, country in zip(axes, COUNTRIES): sub =
df[df['location']==country].dropna(subset=['new_cases']) ax.bar(sub['date'], sub['new_cases'],
color=colors_map[country], alpha=0.35, width=1, label='Daily Cases') ax.plot(sub['date'],
sub['rolling_7d'], color=colors_map[country], linewidth=2, label='7-Day Rolling Avg')
ax.set_title(country, fontweight='bold') ax.legend() plt.tight_layout()
plt.savefig('chart1_country_daily_cases.png', dpi=150) # Overlay: all countries rolling averages on
one axes fig, ax = plt.subplots(figsize=(14, 6)) for country in COUNTRIES: sub =
df[df['location']==country].dropna(subset=['rolling_7d']) ax.plot(sub['date'], sub['rolling_7d'],
linewidth=2.5, label=country, color=colors_map[country]) ax.set_title("7-Day Rolling Average - All
Countries Overlay") ax.legend(); plt.tight_layout() plt.savefig('chart2_rolling_avg_overlay.png',
dpi=150)
```

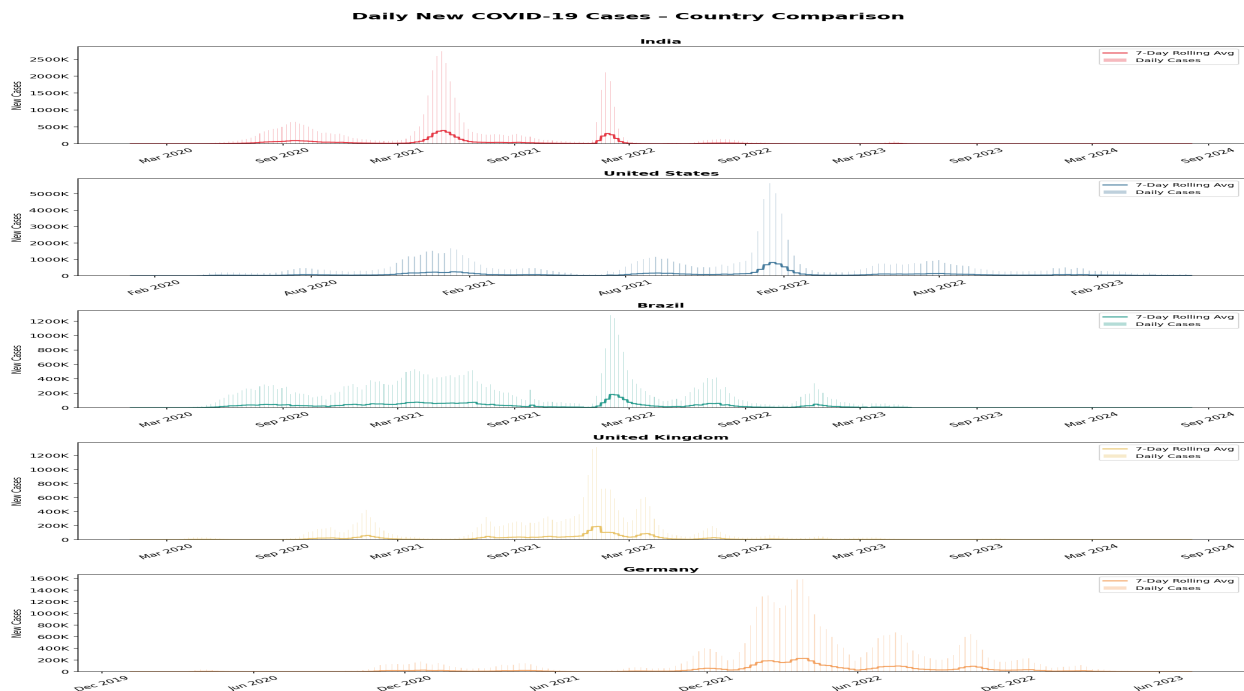


Figure 1 – Daily New Cases (bars) with 7-Day Rolling Average (line) per country

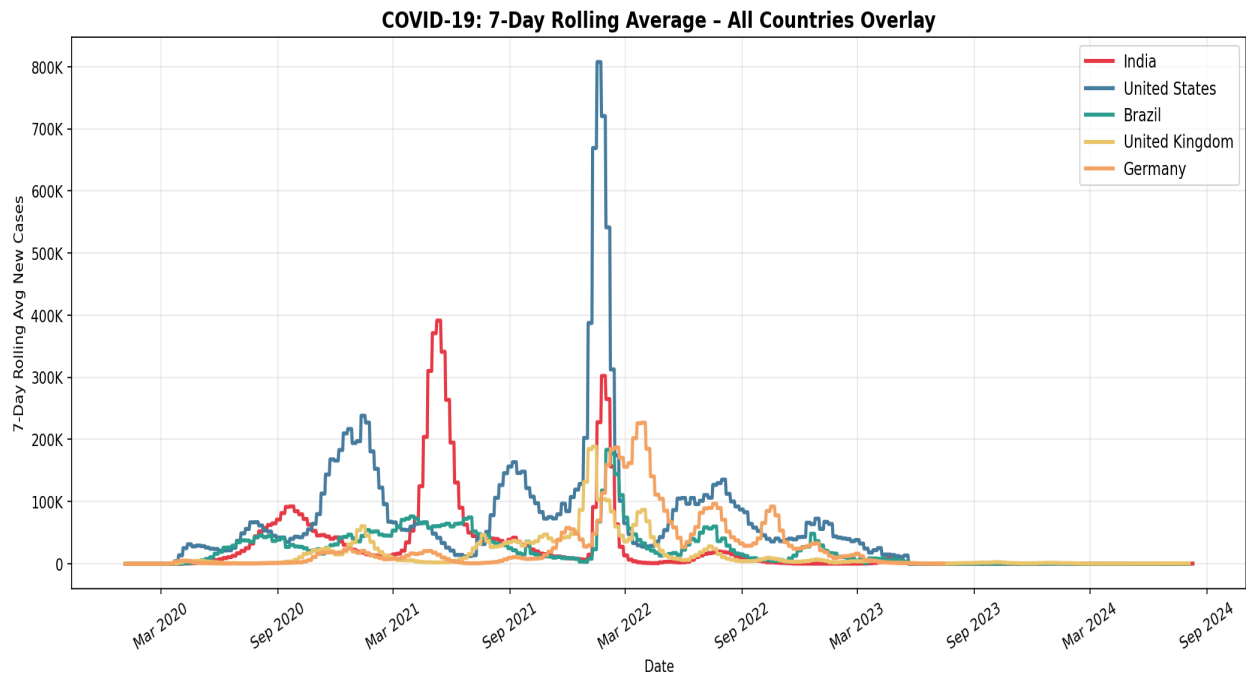


Figure 2 – 7-Day Rolling Average Overlay: All 5 Countries

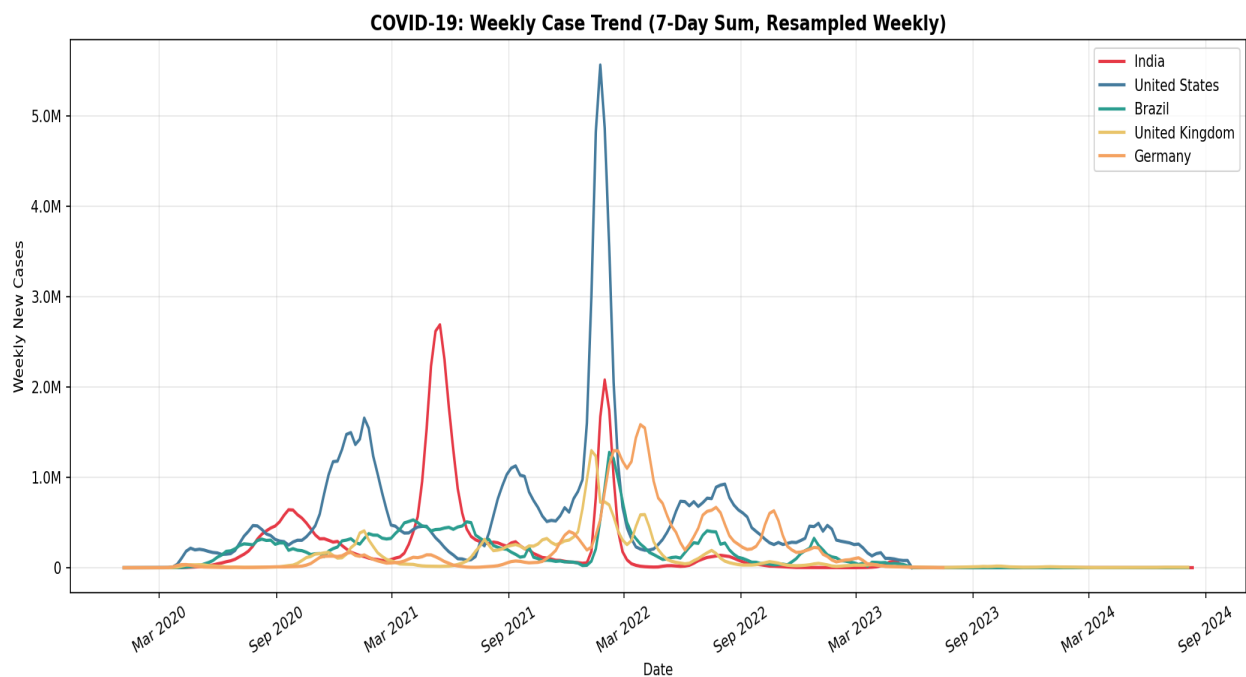


Figure 3 – Weekly Case Trend (7-day sum, resampled weekly)

Interpretation

- **USA** recorded the highest absolute case counts, peaking near 807K cases/day (7d avg) during the Omicron wave (Jan 2022).
- **India** shows a dramatic Delta wave in May 2021 (391K/day), dwarfing its initial Alpha wave.
- **Brazil** experienced 5 distinct waves — the most fragmented trajectory — reflecting inconsistent policy responses.
- **UK & Germany** saw their largest waves during the Omicron period (Jan–Mar 2022).
- The overlay (Fig 2) clearly shows **Omicron's global synchronicity** in early 2022 across all countries.

- Weekly aggregation (Fig 3) confirms the same wave structures at a lower frequency, suitable for policy planning.

3. Peak Detection & Reproduction Rate (R) Insights

SciPy's *find_peaks* algorithm identifies major epidemic waves using prominence and minimum distance constraints. The effective reproduction number R (provided by OWID) quantifies transmissibility — $R > 1$ means exponential growth; $R < 1$ means decline.

Source Code – Peak Detection & R-value Analysis

```
from scipy.signal import find_peaks
peak_results = {}
fig, axes = plt.subplots(5, 1, figsize=(14, 20))
for ax, country in zip(axes, COUNTRIES):
    sub = df[df['location']==country].dropna(subset=['rolling_7d']).reset_index(drop=True)
    vals = sub['rolling_7d'].values
    # Detect peaks: prominence >= 15% of max value, at least 30 days apart
    peaks, props = find_peaks(vals, prominence=vals.max() * 0.15, distance=30)
    ax.plot(sub['date'], vals, color=colors_map[country], linewidth=2)
    ax.scatter([sub.loc[p, 'date'] for p in peaks], vals[peaks], color='red',
               zorder=5, s=60, label='Peaks')
    ax.set_title(f'{country} - {len(peaks)} major waves')
    ax.legend()
# Reproduction rate comparison
fig, ax = plt.subplots(figsize=(14, 5))
for country in COUNTRIES:
    sub = df[df['location']==country].dropna(subset=['reproduction_rate'])
    ax.plot(sub['date'], sub['reproduction_rate'], label=country, color=colors_map[country])
    ax.axhline(1.0, color='black', linestyle='--', linewidth=1.5, label='R=1 threshold')
    ax.set_ylim(0, 3)
    ax.set_title("Effective Reproduction Rate (R) - Country Comparison")
    ax.legend()
plt.savefig('chart5_reproduction_rate.png', dpi=150)
```

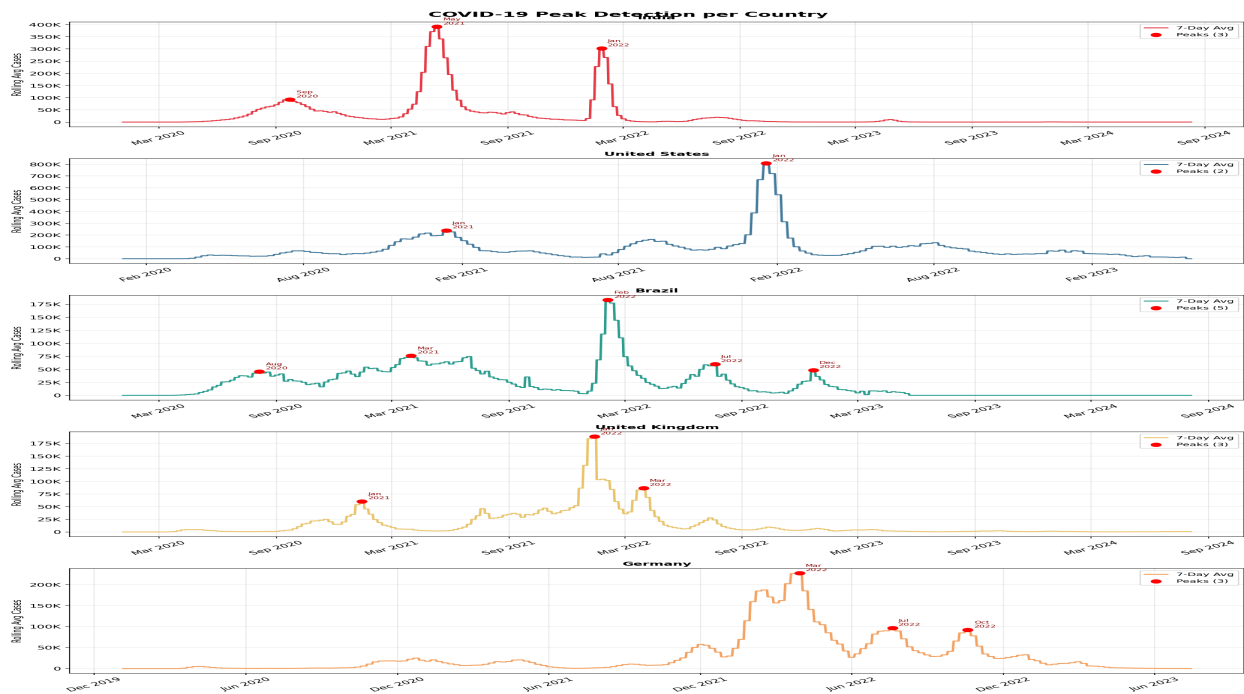


Figure 4 – Peak Detection: Major COVID-19 Waves per Country (Red dots = detected peaks)

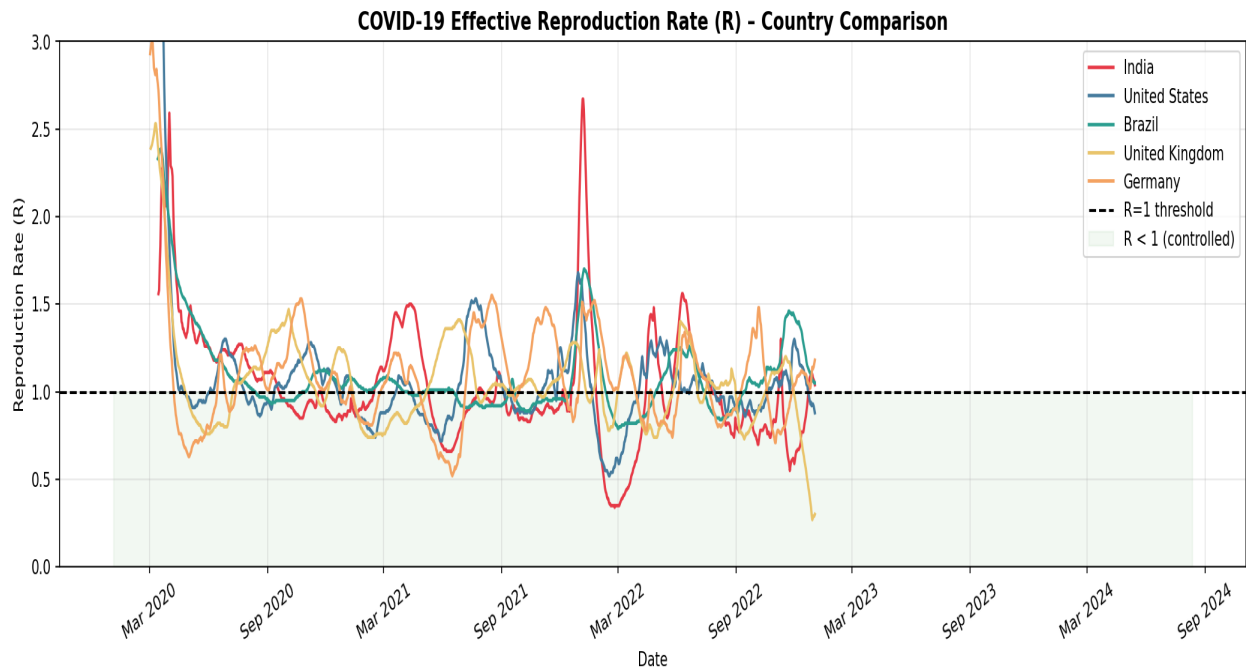


Figure 5 – Effective Reproduction Rate (R): All Countries (dashed = R=1 threshold)

Peak Detection Results

Country	Wave #	Peak Date	Peak Cases/Day (7d avg)
India	1	Sep 2020	92,323
India	2	May 2021	391,280
India	3	Jan 2022	302,157
USA	1	Jan 2021	238,164
USA	2	Jan 2022	807,276
Brazil	1	Aug 2020	45,588
Brazil	2	Mar 2021	76,146
Brazil	3	Feb 2022	183,289
Brazil	4	Jul 2022	59,896
Brazil	5	Dec 2022	48,259
UK	1	Jan 2021	60,382
UK	2	Jan 2022	188,619
UK	3	Mar 2022	86,625
Germany	1	Mar 2022	226,984
Germany	2	Jul 2022	96,447
Germany	3	Oct 2022	92,141

Interpretation

- **India** had 3 waves: Alpha (Sep 2020), Delta (May 2021 — the most deadly), and Omicron (Jan 2022).
- **USA** had 2 detected major peaks: the winter surge (Jan 2021) and the Omicron explosion (807K/day — highest among all).
- **Brazil** had the most waves (5), reflecting prolonged community transmission and delayed vaccination rollout.

- **R > 1 throughout 2020–2022** for all countries confirms persistent transmission before widespread vaccination.
- R dips below 1 during lockdowns and after vaccination campaigns — most visibly in India post-Delta wave.
- The R-value chart shows **synchronised spikes during Omicron** (late 2021 – early 2022) across all 5 nations.

4. Summary Statistics & Conclusions

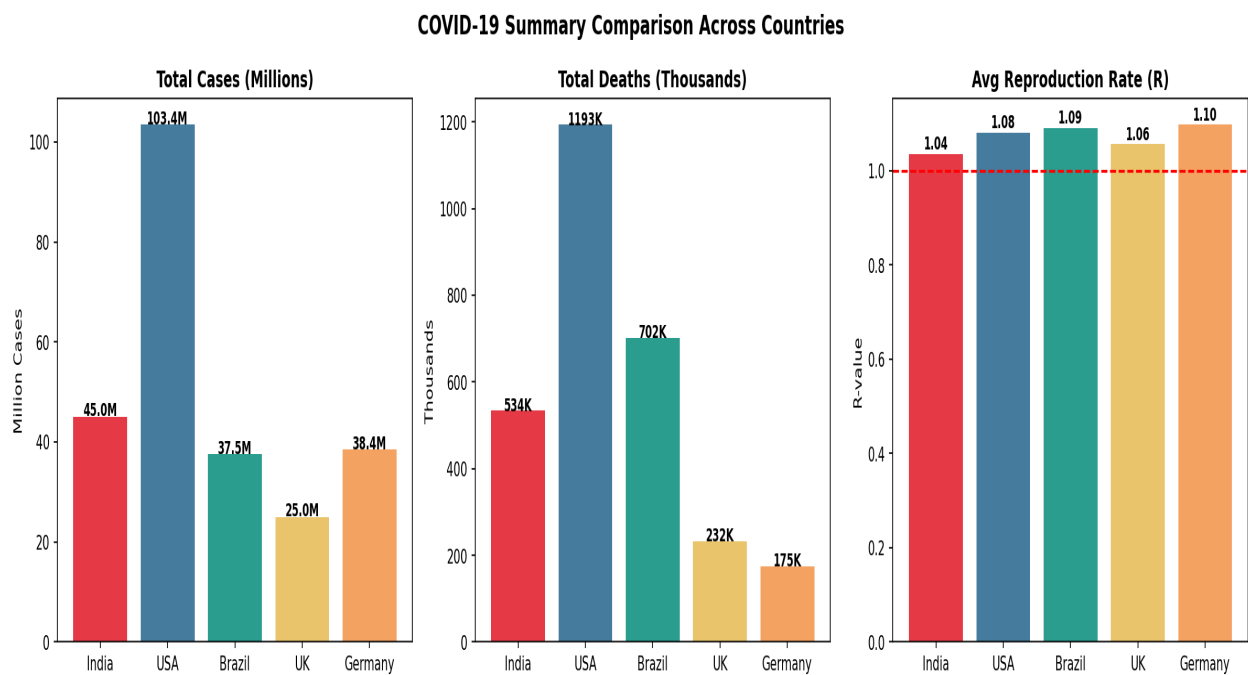


Figure 6 – Summary Comparison: Total Cases, Total Deaths, and Average R-value

Summary Statistics Table

Country	Total Cases	Total Deaths	Peak Daily Cases	CFR (%)	Avg R-value	Waves
India	45.04M	533.6K	2,738,957	1.18%	1.04	3
USA	103.44M	1,193.2K	5,650,933	1.15%	1.08	2
Brazil	37.51M	702.1K	1,283,024	1.87%	1.09	5
UK	24.97M	232.1K	1,320,332	0.93%	1.06	3
Germany	38.44M	175.0K	1,588,891	0.46%	1.10	3

Key Conclusions

- **USA had the highest absolute burden:** 103.44M total cases and 1.19M deaths — heavily driven by the Omicron surge.
- **Germany had the lowest Case Fatality Rate (CFR)** at 0.46%, reflecting strong healthcare infrastructure and early vaccination.
- **Brazil's CFR of 1.87% is the highest** — consistent with healthcare system strain during the P.1 (Gamma) variant surge.
- **India's Delta wave (May 2021)** was the most dramatic single-country peak in proportional terms, with healthcare systems overwhelmed.
- **Average R-values hover near 1.04–1.10** across all countries — indicating sustained transmission over the full period.
- **Omicron (Jan 2022)** was the globally synchronised turning point — highest case counts, but lower CFR due to partial immunity.
- **UK's relatively low CFR (0.93%)** reflects both a robust NHS response and early booster campaign deployment.

- **Rolling averages are essential:** raw daily data is dominated by weekend reporting gaps; 7-day smoothing reveals true epidemiological signal.